Name: Amsalu Tessema Adgeh
Student ID Number: 2026A8000929002
Data Science – Assignment 2

INFERENCE AND UNCERTAINTY: SIMULATION-BASED QUANTIFICATION OF SAMPLING VARIABILITY, POWER AND ASSUMPTION FAILURE

Master seed 20261012 · synthetic data (generator dgp-v1) · Python 3.12.3, NumPy 2.4.4, SciPy 1.17.1

1. Introduction and Pre-Specification

A point estimate answers “how large?” but not “how sure?”. The objective is to quantify the second question with simulation, and to show when the quantification itself fails. The study design below was committed to version control (file *PREREGISTRATION.md*) before any code was executed; the commit order, not the notebook, is the evidence of pre-specification.

Table 1. Pre-specified question, estimand, hypotheses and minimum meaningful effect.

Item	Specification
Question	Does a treatment shift the mean of a continuous outcome relative to control?
Estimand	Δ = E[Y | treated] − E[Y | control], in units of the control standard deviation (average treatment effect under randomisation).
Hypotheses	H_0: $\Delta = 0$ versus H_1: $\Delta \neq 0$; two-sided; $\alpha = 0.05$.
Minimum meaningful effect	$\delta^* = 0.4$ standard deviations; smaller effects are practically negligible regardless of p-value.
Design target	Power of at least 0.80 at $\Delta = \delta^*$.
Decision rule	“Supported” if the 95% interval excludes 0; “compatible with a meaningful effect” if its upper limit is at least δ^*. The interval is always reported.
Assumptions stressed	A1 independence of observations; A2 light tails and variance structure; A3 individual-level randomisation; A4 fixed sample size.

2. Methods

2.1 Estimand, estimator and sampling variance

Let $\bar{y}_T$ and $\bar{y}_C$ denote the treated and control sample means. The estimator of Δ is the observed mean difference

$$D = \bar{y}_T - \bar{y}_C \tag{1}$$

Because the two arms are independent, the covariance term vanishes and

$$Var(D) = \sigma_T^2/n_T + \sigma_C^2/n_C \tag{2}$$

For equal sample size n and equal standard deviation σ per arm, SE(D) = σ√(2/n), and under normality

$$D \sim N(\Delta, 2\sigma^2/n) \tag{3}$$

The standard error therefore decreases proportionally to $n^{-1/2}$: halving the interval width requires four times the data.

2.2 Welch interval and test

With $v_T = s_T^2/n_T$ and $v_C = s_C^2/n_C$, the estimated standard error is $SE = \sqrt{(v_T + v_C)}$ and the test statistic is t = D/SE. Welch (1947) and Satterthwaite (1946) approximate its reference distribution by a t distribution with

$$\nu = (v_T + v_C)^2 / [\, v_T^2/(n_T - 1) + v_C^2/(n_C - 1) \,] \tag{4}$$

degrees of freedom. The (1 − α) confidence interval is

$$D \pm t_{1-\alpha/2,\,\nu} \cdot SE \tag{5}$$

The pooled-t variant instead assumes $\sigma_T = \sigma_C$, using $s_p^2 = [(n_T - 1)s_T^2 + (n_C - 1)s_C^2] / (n_T + n_C - 2)$ and $SE = s_p\sqrt{(1/n_T + 1/n_C)}$.

2.3 Percentile bootstrap

Following Efron (1979), each arm is resampled with replacement from its own observations B times, giving $D^*_b = \bar{y}^*_{T,b} - \bar{y}^*_{C,b}$ for b = 1, …, B. The interval is

$$[\, q_{\alpha/2}(\{D^*_b\}),\, q_{1-\alpha/2}(\{D^*_b\}) \,] \tag{6}$$

where q denotes an empirical quantile. The justification is that the distribution of D* − D given the data approximates that of D − Δ. The method needs adequate sample size, finite variance and independent units, and the bootstrap distribution is centred at the observed D, not at Δ.

2.4 Permutation test

If treated and control share the same distribution, every relabelling of the pooled observations is equally likely. Let D^*_b be the mean difference after a random relabelling. The p-value is

$$p = (\, 1 + \#\{\, b : |D^*_b| \geq |D| \,\}) / (\, B + 1 \,) \tag{7}$$

(Phipson and Smyth, 2010). For cluster-level analysis with m clusters per arm, all C(2m, m) label splits of the cluster means are enumerated (924 splits for m = 6), giving an exact p-value.

Caveat: this procedure tests equality of distributions, not equality of means; with unequal variances and unequal sample sizes it can reject a true H_0: $\Delta = 0$ (Section 4.4).

2.5 Power

For equal n and σ, the t statistic follows a noncentral t distribution with $\nu = 2n - 2$ degrees of freedom and noncentrality

$$\lambda = \Delta / (\sigma\sqrt{(2/n)}) = d\,\sqrt{(n/2)},\ d = \Delta/\sigma \tag{8}$$

so that, with critical value $t_c = t_{1-\alpha/2,\,\nu}$,

$$Power = 1 - F_{nct}(t_c;\ \nu, \lambda) + F_{nct}(-t_c;\ \nu, \lambda) \tag{9}$$

Effect size Δ and noise σ enter only through d: doubling the noise is equivalent to halving the effect. The Monte-Carlo standard error of any simulated proportion is $\sqrt{(p(1 - p)/R)}$ for R replicates.

2.6 Clustered data and the design effect

Suppose m clusters per arm are randomised, each with k subjects and intra-cluster correlation ρ. The variance of a cluster mean is $\sigma^2[\rho + (1 - \rho)/k] = \sigma^2[1 + (k - 1)\rho]/k$, hence

$$Var(D) = 2\sigma^2[1 + (k - 1)\rho]/(mk) = DE \cdot 2\sigma^2/(mk),\ DE = 1 + (k - 1)\rho \tag{10}$$

A naive analysis that treats subjects as independent omits DE, so its z-statistic is inflated by $\sqrt{DE}$ and the actual type-I error of a nominal 5% test is approximately

$$2[\,1 - \Phi(\,1.96 / \sqrt{DE}\,)\,] \tag{11}$$

where Φ is the standard normal distribution function. The valid alternative analyses the m cluster means per arm as the units of observation.

3. Simulation Design and Reproducibility

All randomness derives from one master seed (20261012) split with NumPy's *SeedSequence.spawn* into independent streams, one per section. No external data are used; the baseline process is $Y_T \sim N(\Delta, 1)$ and $Y_C \sim N(0, 1)$. Replicate counts were fixed in advance: 20,000 for sampling distributions, 4,000 per power cell, 2,000 per method-comparison cell (Monte-Carlo SE of a 5% rate ≈ 0.005) with B = 1,000 resamples inside each replicate, and B = 2,000 (bootstrap) and 5,000 (permutation) for the single worked example. Four data-generating scenarios were compared (Table 2).

Table 2. Data-generating scenarios used to stress the nominal 5% procedures.

Scenario	Violated assumption	Construction
S0 baseline	None	Normal(0, 1); 40 vs 40 per arm.
S1 heavy tails	A2 (light tails)	Student-t with 2.5 d.f., rescaled to unit variance; 40 vs 40.
S2 unequal variance and n	A2 (variance structure)	Treated n = 10, σ = 3; control n = 70, σ = 1.
S3 clustered	A1 independence; A3 unit of randomisation	6 clusters per arm × 10 subjects; ICC ρ = 0.3; whole clusters randomised.

4. Results

4.1 Sampling distribution of the mean difference

At n = 40 per arm the simulated standard error is 0.2244 against a theoretical 0.2236 (Table 3). The log-log slope of the standard error against n is -0.501 (theory: −0.5). The bootstrap standard error from a single experiment (0.205) tracks the true value but is centred on that experiment's D (0.438) rather than on Δ (Figure 1c).

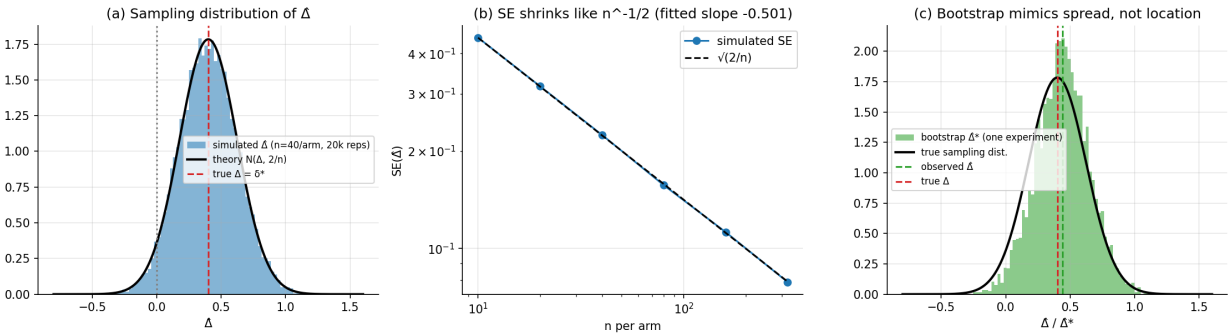


Figure 1. (a) Simulated sampling distribution of D (n = 40 per arm, 20,000 replicates) with the theoretical normal curve; (b) standard error versus n on log-log axes; (c) one experiment's bootstrap distribution compared with the true sampling distribution.

Table 3. Simulated and theoretical standard error of D.

n per arm	Mean of D	Simulated SE	Theoretical √(2/n)	Relative error (%)
10	0.4069	0.4475	0.4472	+0.06
20	0.3982	0.3170	0.3162	+0.23
40	0.4003	0.2244	0.2236	+0.35
80	0.4003	0.1568	0.1581	-0.84
160	0.4000	0.1124	0.1118	+0.50
320	0.4004	0.0788	0.0791	-0.34

4.2 Worked example: intervals and tests

A single experiment with n = 40 per arm and true Δ = 0.4 (the first draw from a frozen seed, not selected) gave D = 0.438. The Welch 95% interval is [0.023, 0.854] (p = 0.0391); the percentile-bootstrap interval is [0.019, 0.834] (B = 2,000); and the permutation p-value is 0.0400 (B = 5,000). By the pre-declared rule the result is classified as “supported & compatible with ≥MME”. The lower limit lies close to zero because the design has a power of only 0.42. Moreover, among statistically significant results the average |D| is 2.01, 1.51 and 1.12 times the true effect at n = 20, 40 and 100 per arm, respectively (Type-M error; Gelman and Carlin, 2014): a bare point estimate from an under-powered study is systematically exaggerated.

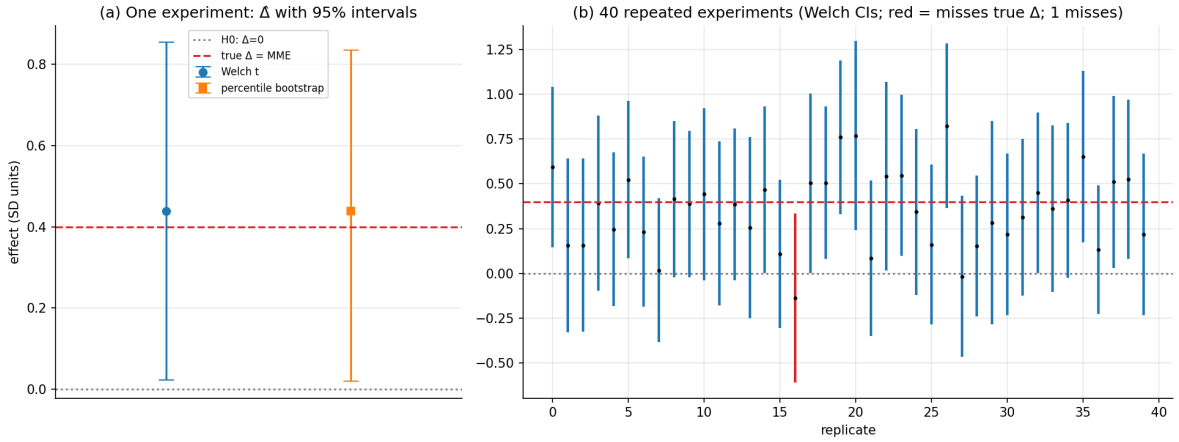


Figure 2. (a) The worked example with Welch and bootstrap 95% intervals relative to H_0 and the true effect; (b) 40 repeated experiments (red intervals miss the true Δ).

4.3 Power as a function of sample size, noise and effect

Power was simulated on a grid of 8 sample sizes, 5 effects and 3 noise levels with 4,000 experiments per cell. Simulated power agrees with exact noncentral-t power (mean absolute difference 0.0035, maximum 0.0166 over 120 cells; Monte-Carlo SE at most 0.008), and the mean type-I rate across null cells is 0.0492. The smallest sample for 80% power at the minimum meaningful effect ($\sigma = 1$) is 100 per arm (simulated power there: 0.803), whereas the worked example used 40. Cells sharing the same $d = \Delta/\sigma$ differ in simulated power by at most 0.024, confirming that effect and noise act only through d (Figure 3c).

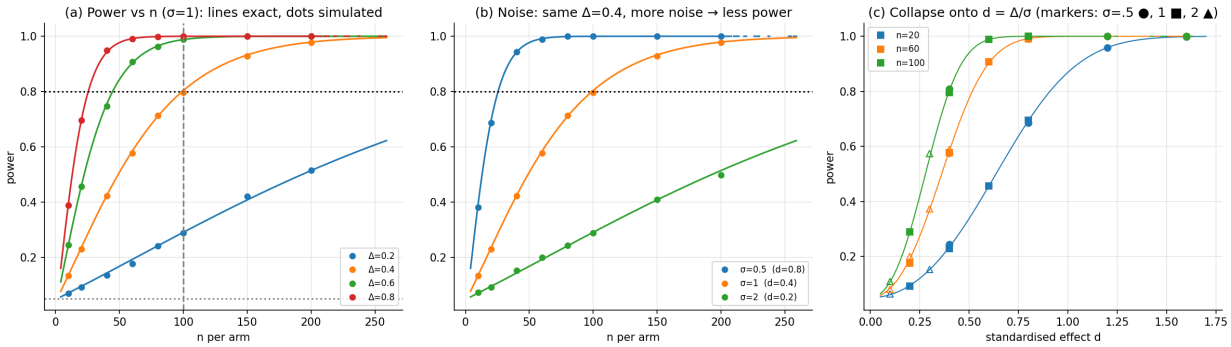


Figure 3. (a) Power versus n for several effects (lines: exact; points: simulated); (b) the same effect with increasing noise; (c) collapse of all (Δ, σ) pairs onto the standardised effect d.

4.4 Counterexample: violating independence

Table 4 reports, for every scenario and method, the rejection rate when $\Delta = 0$ (type-I error; nominal 0.05), the rejection rate when $\Delta = \delta^*$ (power) and the coverage of the 95% interval at $\Delta = \delta^*$ (nominal 0.95).

Table 4. Operating characteristics by scenario and method (2,000 replicates per cell). Cluster-aware methods apply only to S3.

Scenario	Method	Type-I	Power	Coverage
S0 baseline	Welch t	0.041	0.414	0.949
S0 baseline	Pooled t	0.041	0.415	0.949
S0 baseline	Percentile bootstrap	0.051	0.445	0.942
S0 baseline	Permutation (mean diff)	0.042	0.416	–
S1 heavy tails	Welch t	0.045	0.588	0.963
S1 heavy tails	Pooled t	0.045	0.588	0.962
S1 heavy tails	Percentile bootstrap	0.073	0.613	0.935
S1 heavy tails	Permutation (mean diff)	0.050	0.600	–
S2 unequal var & n	Welch t	0.051	0.063	0.957
S2 unequal var & n	Pooled t	0.355	0.351	0.696
S2 unequal var & n	Percentile bootstrap	0.101	0.110	0.905
S2 unequal var & n	Permutation (mean diff)	0.350	0.344	–
S3 clustered	Welch t	0.299	0.580	0.705
S3 clustered	Pooled t	0.299	0.580	0.705
S3 clustered	Percentile bootstrap	0.314	0.589	0.695
S3 clustered	Permutation (mean diff)	0.302	0.579	–
S3 clustered	Cluster-mean Welch t	0.036	0.163	0.953
S3 clustered	Cluster bootstrap	0.102	0.289	0.898
S3 clustered	Cluster permutation (exact)	0.041	0.178	–

In S3 the design effect is DE = 3.70, which predicts a naive type-I rate of 0.308. Across 20,000 clustered null experiments the observed rate is 0.315. The naive analysis believes the standard error is 0.178, whereas the true standard deviation of D is 0.349 (ratio 1.97; $\sqrt{DE} = 1.92$). Consequently its 95% intervals cover the true null value only 68.5% of the time. A cluster-aware analysis restores validity (type-I 0.046). Of the 6,309 naive “discoveries” under a true null, 85.4% are not significant under the correct analysis, so the substantive conclusion changes in most such cases.

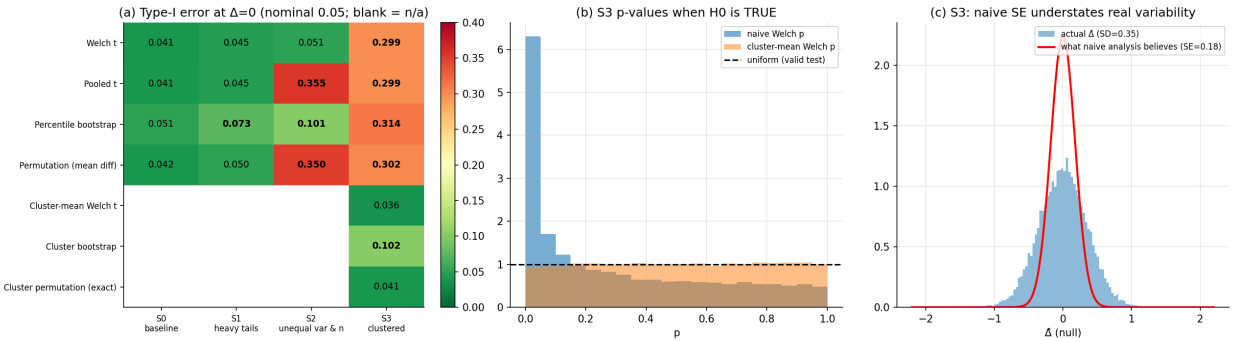


Figure 4. (a) Type-I error by scenario and method (blank: not applicable); (b) distribution of p-values in S3 when H_0 is true; (c) actual sampling distribution of D in S3 compared with what the naive analysis assumes.

Two single datasets (true $\Delta = 0$) illustrate this. In the first replicate with naive $p < 0.01$ (replicate 0), the naive interval is [0.390, 1.065] ($p = 0.0000$) and the cluster-aware interval is [0.034, 1.421] ($p = 0.0415$); *the verdict did not change* here, although the interval became twice as wide. This negative instance is retained deliberately. In a representative case chosen by a stated rule (first replicate with naive $p < 0.05$ and cluster-aware $p \geq 0.05$; replicate 2; this pattern covers 85% of naive discoveries), the naive interval [0.084, 0.776] ($p = 0.0153$) is "supported & compatible with ≥MME", while the cluster-aware interval [-0.291, 1.151] ($p = 0.2063$) is "inconclusive/negligible".

5. Discussion, Negative Results and Limitations

Evidence that challenges a simple narrative is reported rather than removed:

(1) The cluster bootstrap is not a remedy when clusters are few: with 6 clusters per arm its type-I rate is 0.102, versus 0.036 for the cluster-mean t test and 0.041 for the exact cluster permutation test.

(2) Validity costs power: the cluster-aware test has power 0.163 at the minimum meaningful effect, against 0.580 for the invalid naive test. With DE = 3.70 the design is badly under-powered.

(3) The permutation test is not a test of means. Under unequal variances and sample sizes (S2) its type-I rate is 0.350, comparable to the pooled t test (0.355), while Welch's test holds 0.051.

(4) Heavy tails did not reduce Welch power at fixed variance (power 0.588 versus 0.414 at baseline). The median sample standard deviation was 0.79 against a population value of 1, which is consistent with, but does not prove, typical samples underestimating the standard deviation; the cost appears instead as a bootstrap type-I rate of 0.073. Efficiency relative to

rank-based tests was not studied.

(5) The baseline Welch type-I rate in the 2,000-replicate comparison (0.041) lies about 2.3 Monte-Carlo standard errors below 0.05, whereas the independent 4,000-replicate grid at n = 40 gave 0.056 and the grid-wide mean is 0.049. This is interpreted as Monte-Carlo noise and was not “corrected” by re-seeding.

(6) Conclusions are specific to the simulated settings (one effect size, one ICC, one cluster layout, normal outcomes except S1). No real data were analysed, and no multiplicity adjustment is applied because the grid is descriptive rather than inferential.

6. Verification and Reproducibility

Thirteen automated cross-checks were executed inside the notebook (Table 5): hand-written estimators against SciPy, simulation against closed-form theory, and the clustered type-I rate against the design-effect prediction. Failed checks: 0. A seed re-draw reproduced the data bit-for-bit (True), and a second full execution reproduced all tables and figures byte-for-byte. Dependencies are pinned in *requirements.txt*; artefact hashes are stored in *results/manifest_sha256.json*; the full run history, including a superseded first run and the errors found in it, is retained in the repository.

Table 5. Executed verification checks.

Check	Value	Result
V1 Welch p vs scipy.ttest_ind	6.939e-17	PASS
V1b Welch CI limits vs scipy	1.11e-16	PASS
V2 bootstrap CI vs scipy.stats.bootstrap (B=20000 each)	0.006186	PASS
V3 permutation p vs scipy.stats.permutation_test	0.00275	PASS
V3b exact permutation returns p=1 for zero effect data	0	PASS
V5 SE scaling slope on log-log (theory -0.5)	-0.501	PASS
V5b max relative SE error vs sqrt(2/n), %	0.8376	PASS
V4 max |sim − exact| power over 120 cells	0.01658	PASS
V4b mean |sim − exact|	0.003474	PASS
V6 baseline type-I rate (mean over null cells)	0.04917	PASS
V7 naive S3 type-I vs DE-based prediction	0.007226	PASS
V7b variance inflation ≈ DE	0.03642	PASS
V8 S1 power: own Welch vs scipy.ttest_ind on identical data	0	PASS